

Apache Hadoop based effective sentiment analysis on demonetization and covid-19 tweets

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ABSTRACT

In Recent, Twitter is the well-known public Network acquires a huge number of tweets. Sentiment analysis in twitter data are tremendously valuable in social media observing as it allows getting an overview of extensive global opinion in certain issue. This data are utilized for industrial, government, social and economic approaches by analyzing the tweets as per the requirement of the user. Processing and storing these data are more complicated to analyze. Hadoop is a distributed environment which process with Big and Huge variety of dataset which supports processing components that collectively called Hadoop Ecosystem. In this paper, regular tweets are analyzed by sentiment analysis technique in Hadoop Eco system. Dataset are taken from Kaggle data repository. This research has done by Apache Pig in Demonetization and Covid 19 twitter dataset.

1. Introduction

Social Media like twitter and LinkedIn are very influential communication tools. Thousands of tweets are posted regularly in these social networks [1–3]. The improvement in the technology has made to deliver their perception and depends on these reviews [4–6]; user can send opinions to make decisions [7,8]. The data taken from tweeter can be used for analyzing people's judgment [9–14]. The data stored from these social media can be analyzed the opinion and sentiments of the user. Sentiment Analysis is the process of user's sentiments as positive or negative estimations. In first, it can be identified as subjective or objective. If the statement is subjective, it will be divided as positive or negative opinions [10].

For analyzing the large number of data, Hadoop is suitable for large file processing and large storage of wider data units. Generally, big data processing is a critical task to recognize the originality of the data. Most of the data are utilized in social network are not structured [22–26]. That is 79% of data which used in social media are unstructured. Therefore, data process becomes complex task in big data Analytics. Apache Hadoop is an efficient and convenient framework to analyze the social media content [11]. In this paper, Apache pig is used to analyze the twitter dataset [12]. Using the Apache Flume, the demonetization and covid19 tweets were collected from web server and aggregated it into HDFS for Apache Pig Analysis. Apache Flume is distributed open source tool for aggregating log activities from the web server into Hadoop Distributed File Systems (HDFS) [13].

Unstructured data processing become crucial to analyzes and predicts the accuracy of the data as it is not fit into the fixed size [27–32]. And also social media data like twitter are having emotions, opinions, and user's judgement and meaningless symbols as reviews. These reviews are used for advertising a particular product and it can be utilized to know the quality of the product. [14] Moreover, Tweets in social media have made some decisions for the important government plans and Announcements [15].

1.1. Problem statement

As discussed earlier, twitter data are not comfortable to process in a simple way as they have not predetermined in a particular structure. So this leads to the problem for researchers who want to access the data. They could not able to fetch the data in their way because every tweet contains emotions, meaningless words, symbols for cheerful, miserable and personal belief. And also user's tweets have some symbols. The above known problems are overcome by Hadoop core components that are perfectly suitable to deals with the problem of collecting data from web server, storing wider range of data in HDFS and effective processing of the data by using Hadoop Tools. The main intend of this paper is to prove the significance of sentiment analysis and how it is handled by the Effective techniques of Big data Analytics. In this paper Twitter Sentiment Analysis (TSA) algorithm is implemented in demonetization and covid -19 datasets for analyzing tweets, then their result is compared by using classification algorithms.

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1.2. Organization of the paper

This paper is prepared as follows: Section-2 includes the Related work of this twitter data analysis. Section-3 undertakes proposed research method of twitter analysis and Section- 4 discusses the empirical results. Conclusion of the proposed method and scope for further research work discussed in section-5.

2. Related work

In this section, the related work for this paper can be divided into three sections. In first collecting data from social media is complex task done by Aache Flume.

2.1. Collecting tweets

Review of literature pointed out fetching tweets from Social media. Battineni, G., et al. has carried out a research for executing sentiment analysis which impacting online suggestions affects quality and reliability of health. [11]. Hadoop is the distributed computing environment which consists of three layers. 1. Application layer, 2. Resource allocation layer (YARN), 3. Storage layer. These three layers are used to process the client request, allocating job to their task trackers and to store the data as blocks into HDFS. Twitter data collection is a process of collecting tweets of information collected from external data source [16]. After collecting the data, it pre-processed by five different phases like Tokenization, Stemming, and Case Conversion Special character removal from the tweets, and stop word removal.

2.2. Data processing

The processed data can be uploaded and stored into HDFS. El Alaoui, I. et al have discussed the quality of big data analytics on social networks. In this research, the actual domain of big data process discussed well. [15]. Samuel et al presented a new sentiment analysis in his research article Public Sentiment Insights and Machine Learning for Tweets Classification. Information [17]. Dhanya, N. M et al have proposed a Sentiment analysis on tweeter datasets by using algorithm Machine learning algorithm techniques [18]. Chakraborty et al. [19] have presented a [Sentiment Analysis on Movie Reviews Using Deep Learning Techniques](#) by Gaussian filter based fuzzy rule base algorithm is used for analyzing the sentiment from tweets which yields 79% accuracy. Prayag Tiwari set al have proposed a [Sentiment Analysis for air way on Twitter Data](#) In this research, twitter data analyzed by association rule mining algorithm and BIRCH clustering methods. The study obtained a number of positive, negative sentiment, and their association regarding consumer tweets. From this work, it is proving that positive tweets associated with the words of tweets liked by the passengers of air ways [20]. Sharma, Ankita, et al. have proposed Sentimental analysis on twitter data regarding elections [21] R Language is worn for the data acquisition, data pre-processing, and sentiment analysis is performed by the Tweets which were collected from the phase of Jan 2019 to March 2019. to gain the opinion polarity.

3. Proposed method

In this paper, this research work consists of five phases, Pre-process of data, Feature selection from the dataset, and Upload the pre-processed data to HDFS, Sentiment Analysis of Data and classifying their polarity and evaluating outcomes using classifiers. The architecture of the proposed method is shown in the [fig. 1](#). The basic functions of preprocessing techniques are shown in the [fig. 2](#).

3.1. Dataset description

In demonization dataset, 14940 data instances and 15 attributes related to tweets. This paper examines the tweets of COVID-19 from Jan-

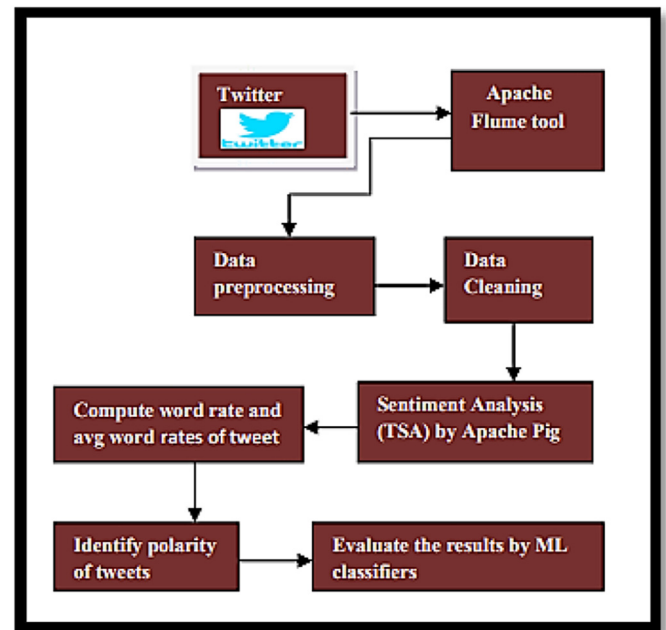


Fig. 1. Architecture of the Proposed Method.

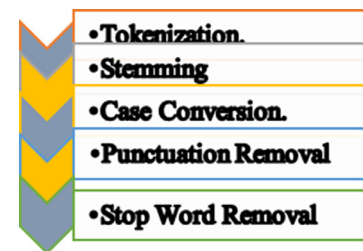


Fig. 2. The Preprocessing Methods

Table 1
Pre-processing Algorithm

Algorithm1 Pre-process (DS, PD)
// Input D: Twitter dataset
// Output RD: pre-processed dataset
(1) Select the Original Training Dataset D
(2) Remove the unwanted and noisy data from tweets
(3) Tweets labeled with sentiments
(4) Return The Preprocessed Dataset RD

uary 2020 to May 2020 in India and its affect. In this dataset, tweets are categorized into 3 groups (Positive, Negative and Neutral). This study collects data on social media from early phase of COVID-19 spread.

3.2. Pre-processing methods

The twitter data is not fitting to analysis. So, the data have been pre-processed by Algorithm1 is shown in the [Table 1](#).

Tokenization is the process of breaking a stream of sentences into a collection of words. In Second, the process of eliminating words to their stem is called stemming. The same word can be used in a different synonym for grammatical purpose such as Search, Searches Searching. Case Conversion phase changes the text into whichever the lowercase or the uppercase. Punctuation Removal process is removing Punctuations in a text because they do not give any information and remove the punctuation symbols from the word. Stop Word Removal is utilized to Stop words consist of prepositions, help verbs, articles etc. After pre-processing the data, the resultant dataset uploaded into HDFS. Data are

Table 2
Twitter Sentiment Analysis (TSA) Algorithm.

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Algorithm 2: Twitter Sentiment Analysis (TSA)
//Input: dataset: no of tweets, dictionary words and their rate
//dwr: dictionary_word_rate, Twt:Tweets, Cs:Corpus,
//Avg_R:Average rate
//Output: categorization: positive, negative or neutral
1. Load the dataset into Pig-x local mode using pig storage.
2. Extract id and Twt
3. divide Twt text into words
4. Load Dictionary and join token and dictionary
5. For each attribute Twt in Cs do
5.1. While(words) do
5.2. If word = any word in dwr then
5.3. wr = dwr;
5.4. Continue;
5.5. End
5.6. End
6. Avg_R = Avg(wr)
6.1. If Avg_R ≥ 0.0 then
6.2. Tweet is positive
6.3. Else if Avg_R < 0.0 then
6.4. Tweet is negative
6.5. Else neutral
6.6. End_if
7. Return Avg_R
8. Resultant dataset is applied into three classification models
9. Perform 10 folds cross validation
10. Analyze and compare the results.

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categorized into positive, negative. The unrelated tweets are also declared as neutral in demonetization and Covid-19 dataset. After uploading the dataset into HDFS, the dataset is analyzed by Apache Pig. [Table 2](#)

3.3. Sentiment analysis

A dataset with the positive, negative and neutral label created as training dataset. The data is pre-processed to remove some of the unnecessary attributes from the dataset. An assortment of pre-processing steps such as removing punctuations, emoticons, twitter specific terms and stemming are done and the data is cleaned for analysis. Thus the data is fitting for all the learning algorithms. [Fig. 3](#) shows that positive tweets are larger than the negative tweets.

$$\text{Average Rate (Avg_R)} = \frac{\text{Sum of Word rate}}{(\text{No of Words in Tweet})} \quad (1)$$

After the process of tokenization, compute the rating of tweet id, tweet for categorization of Tweets. Calculated value is utilized to classify the tweets into positive and negative are shown in the [Fig. 3](#).

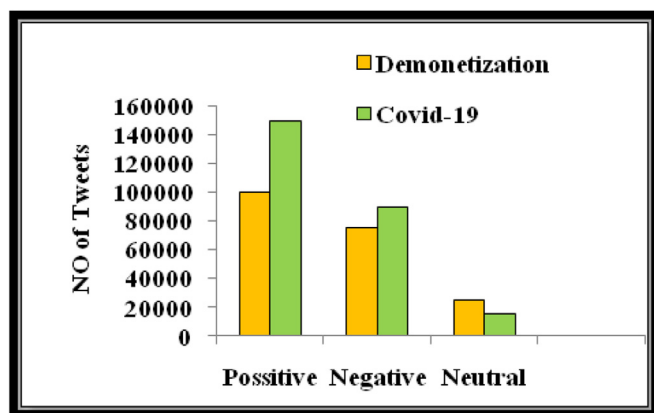


Fig. 3. Sentiment Analysis of Twitter data
a. Covid19 Tweets from Jan to May 2020

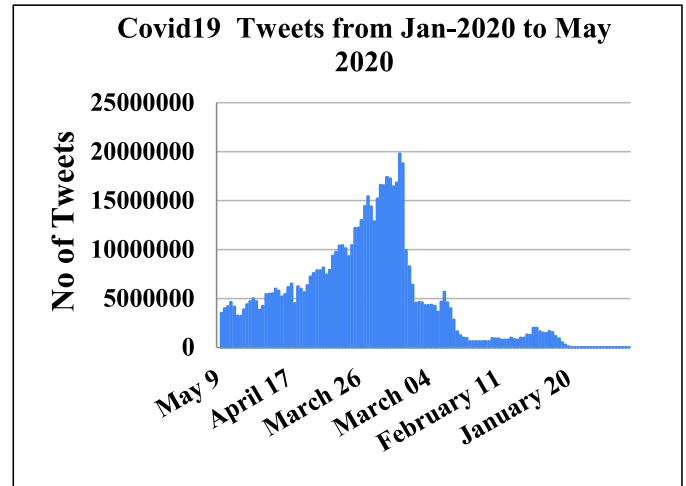


Fig. 3. Continued

Table 3
Performance Analysis of Covid-19 dataset

Classification Methods	Accuracy (%)	Precision (%)	Recall (%)
SVM	70.66	71.45	71.33
NB	66.97	67.31	67.33
RNN	69.34	70.04	70.05

Table 4
Performance Analysis of Demonetization dataset

Classification Methods	Accuracy (%)	Precision (%)	Recall (%)
SVM	92.25	93.43	93.40
NB	73.00	74.76	74.77
RNN	87.00	87.97	87.92

4. Results

The data were cleaned, pre-processed, and annotated into three categories positive, negative, and neutral. In Hadoop eco system, apache pig and apache flume are used. After categorized the dataset, three Machine learning techniques such as support vector machine (SVM), Naive bayes (NB) and Recurrent neural networks (RNN) were chosen for performing the classification task. Classification is the process that maps every tuple in the dataset to make resultant variables. The output variables are said to be a class labels. The mapping algorithm forecasts the class label for every tuple from the given dataset. The performance analysis of the classifiers is examined by accurate prediction of class variable for the test data. The results of the classification process of Covid 19 dataset are shown in the [Table 3](#) and demonetization datasets are shown in the [Table 4](#). Performance evaluators such as Accuracy, Precision, Recall, F-Measure, ROC are calculated and compared with all the three classifiers SVM, NB, RNN are graphically Represented in the [Figs. 4,5,6,7,8,9](#). The SVM classifier produced the highest accuracy of 98.25% in demonetization and 70.66% in Covid -19 dataset then RNN and Naïve Bayes. The average performance and error rate are computed for all 10-fold cross validation of two datasets as shown in the [Fig. 5](#). [Table 5](#) shows the F-Measure and ROC area of Two datasets are rendered. According to the classifiers, SVM and RNN is outperformed in both demonetization and covid -19 dataset. The [Table 6](#) shows the comparison with the state of art literature.

5. Conclusion

Twitter Sentiment analysis (TSA) algorithm is used to identify the polarity of the Twitter dataset. TSA algorithm has been applied on de-

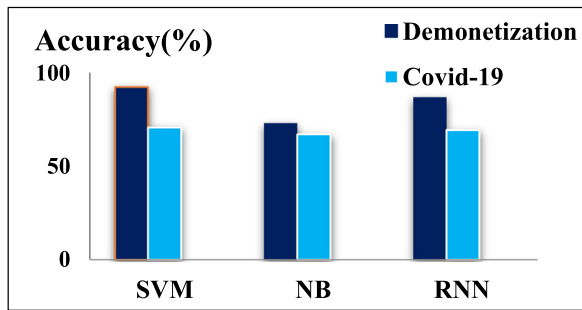


Fig. 4. Performance Analysis of classifiers Accuracy (%)

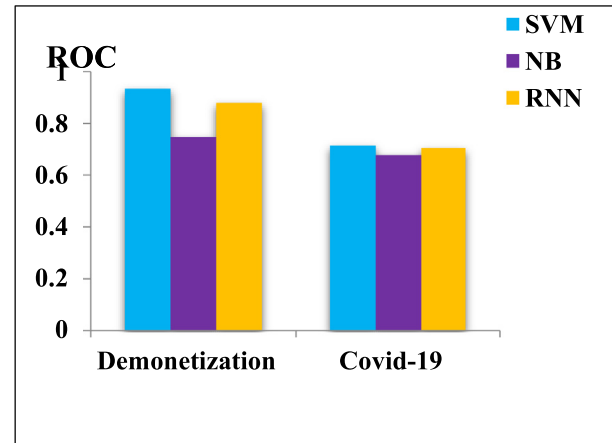


Fig. 8. Comparative ROC analysis of TSA Algorithm using different Dataset

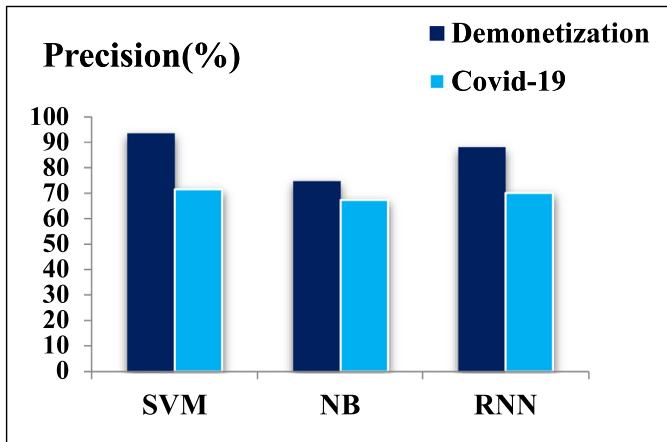


Fig. 5. Performance Analysis of classifiers Precision (%)

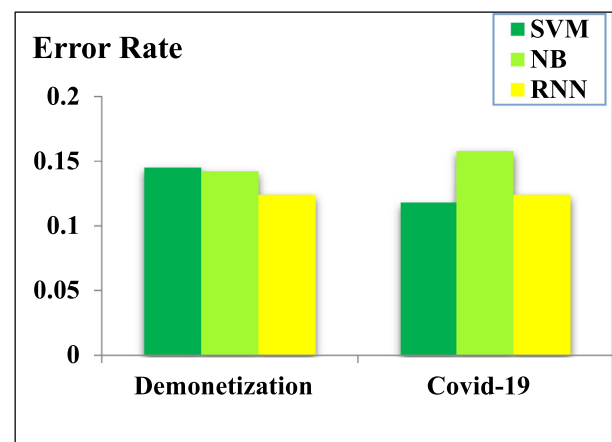


Fig. 9. Error rates of TSA Algorithm using different Dataset

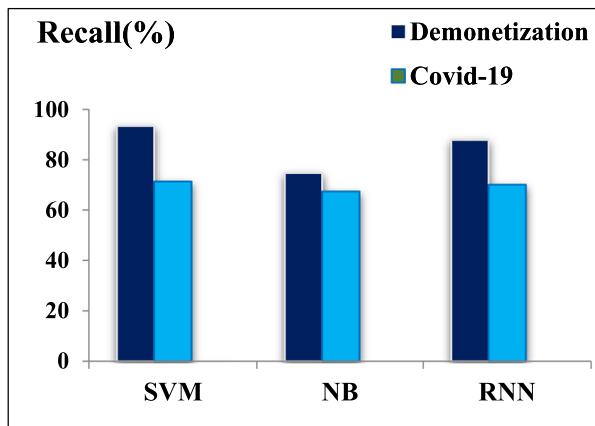


Fig. 6. Performance Analysis of classifiers Recall (%)

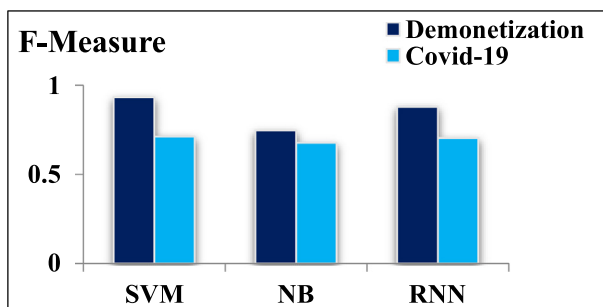


Fig. 7. Performance Analysis of classifiers F-Measure (%)

Table 5

F-Measure and ROC for the classification methods.

Classification Methods	Demonetization		Covid-19	
	F-Measure (%)	ROC Area	F-Measure (%)	ROC Area
SVM	0.93	0.93	0.71	0.71
NB	0.75	0.75	0.68	0.68
RNN	0.88	0.88	0.70	0.70

Table 6

comparison of state of art Literature survey with proposed system

Sl.no.	Method	Accuracy (%)
1	Proposed Method(TSA)-Demonetization	92%
2	Proposed Method(TSA)- Covid 19	70.6%
3	Nalini Chintalapudi et al(2021)	89%
4	Gopi Battineni et al (2021)	96%
5	Sarra Hasiu et al(2021)	81.7%
6	Divya Sehgal et al(2016)	72.22%

monetization and covid-19 dataset. The analyses of the positive and negative tweets of two datasets are identified and the results are evaluated by classifiers such as Support vector machine(SVM), Naive bayes(NB) and Recurrent neural network (RNN) algorithms. SVM is outperformed and gained more accuracy than NB and RNN. RNN classifiers took minimum execution time than other classifiers. From this study, Apache Pig is the best tool for analyzing large number of instances in short execution time. SVM and RNN are suitable classifiers for sentiment Analysis

on twitter dataset. In this work, regional language like Hindi, Tamil are eliminated and it considered as neutral. As the further enhancement of the proposed method, all regional Languages (words) will be considered for Sentiment analysis to increase more accuracy. Considering the merits and demerits of the proposed system.

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